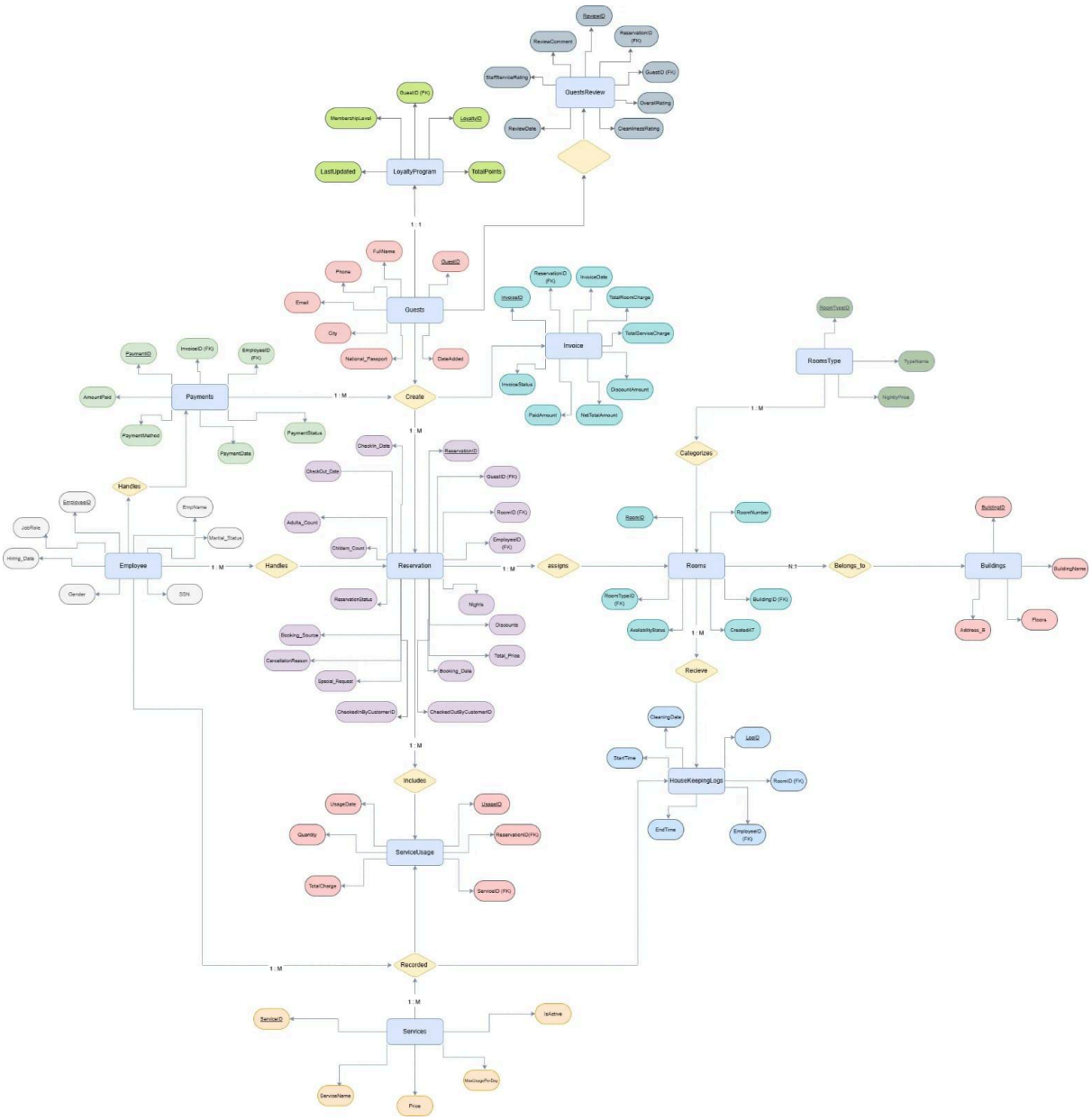
ReviewID
ReviewComment
ReservationID (FK)
StaffServiceRating
GuestsReview
GuestID (FK)
OverallRating
ReviewDate
CleanlinessRating
GuestID (FK)
MembershipLevel
LoyaltyID
LastUpdated
LoyaltyProgram
TotalPoints
1 : 1
FullName
GuestID
Phone
Email
Guests
City
National_Passport
DateAdded
ReservationID (FK)
InvoiceDate
InvoiceID
TotalRoomCharge
Invoice
TotalServiceCharge
InvoiceStatus
DiscountAmount
PaidAmount
NetTotalAmount
RoomTypeID
RoomsType
TypeName
NightlyPrice
PaymentID
InvoiceID (FK)
EmployeeID (FK)
AmountPaid
Payments
1 : M
Create
PaymentMethod
PaymentDate
PaymentStatus
1 : M
Checkin_Date
ReservationID
CheckOut_Date
GuestID (FK)
Handles
EmployeeID
EmpName
RoomID (FK)
Adults_Count
JobRole
Marital_Status
EmployeeID (FK)
Children_Count
Hiring_Date
Employee
1 : M
Handles
Reservation
1 : M
assigns
Rooms
N:1
Belongs_to
Buildings
BuildingID
BuildingName
Gender
SSN
ReservationStatus
Nights
Discounts
Booking_Source
Total_Price
CancellationReason
Special_Request
Booking_Data
CheckedInBy_CustomerID
CheckedOutBy_CustomerID
Categorizes
RoomID
RoomNumber
RoomTypeID (FK)
BuildingID (FK)
AvailabilityStatus
CreatedAT
1 : M
Recieve
Address_B
Floors
1 : M
Includes
ClearingDate
LogID
StartTime
HouseKeepingLogs
RoomID (FK)
EndTime
EmployeeID (FK)
UsageDate
UsageID
Quantity
ReservationID(FK)
ServiceUsage
TotalCharge
ServiceID (FK)
1 : M
Recorded
1 : M
ServiceID
Services
IsActive
ServiceName
Price
MaxUsagePerDay

Quastions .s...\Admin (54))* | LAST Q.sql - not connected

```sql
-- ============================================================
-- Q2: Room Availability View (by date range)
-- ============================================================
-- Shows every room with its type and nightly rate.
-- AvailabilityStatus in the view is the physical status from Rooms.
-- To check availability for a date range we filter in the query below.

CREATE VIEW vw_RoomAvailability AS
SELECT
    ro.RoomID,
    ro.RoomNumber,
    rt.TypeName          AS RoomType,
    rt.NightlyPrice      AS NightlyRate,
    ro.AvailabilityStatus,
    ro.BuildingID,

    -- Attach any active/future reservation dates so we can filter later
    r.ReservationID,
    r.check_in_date,
    r.check_out_date,
    r.reservation_status

```

110 % | 10 | 0 | Ln: 585, Ch: 21 | MIXED | CRLF | UTF-8

Results | Messages

	RoomID	RoomNumber	RoomType	NightlyRate	BuildingID
1	R3	103	Double	900.00	B1
2	R4	104	Double	900.00	B1
3	R8	108	Double	900.00	B1
4	R9	109	Double	900.00	B1
5	R13	203	Double	900.00	B2
6	R14	204	Double	900.00	B2
7	R18	208	Double	900.00	B2
8	R19	209	Double	900.00	B2
9	R23	303	Double	900.00	B1
10	R24	304	Double	900.00	B1
11	R28	403	Double	900.00	B2
12	R29	404	Double	900.00	B2
13	R1	101	Single	500.00	B1
14	R2	102	Single	500.00	B1
15	R6	106	Single	500.00	B1
16	R7	107	Single	500.00	B1
17	R11	201	Single	500.00	B2
18	R12	202	Single	500.00	B2
19	R16	206	Single	500.00	B2

Quastions .s...\Admin (54))* | LAST Q.sql - not connected

```sql
-- ============================================================
-- Q3: Daily Occupancy Rate
-- ============================================================
-- Generates one row per calendar date in the range.
-- OccupiedRooms = rooms that have an active (checked_in / confirmed)
-- reservation covering that specific date.

DECLARE @RangeStart DATE = '2026-01-01';
DECLARE @RangeEnd   DATE = '2026-03-17';

-- Build a calendar CTE for the date range
WITH Calendar AS (
    SELECT @RangeStart AS CalDate
    UNION ALL
    SELECT DATEADD(day, 1, CalDate)
    FROM   Calendar
    WHERE  CalDate < @RangeEnd
),
-- Total active rooms (not under maintenance)
RoomInventory AS (
    SELECT COUNT(*) AS TotalRooms
```

110 % | 10 | 0 | Ln: 634, Ch: 19 | MIXED | CRLF | UTF-8

Results | Messages

	Date	TotalRooms	OccupiedRooms	OccupancyRate%
1	2026-01-01	28	1	3.57
2	2026-01-02	28	1	3.57
3	2026-01-03	28	1	3.57
4	2026-01-04	28	1	3.57
5	2026-01-05	28	1	3.57
6	2026-01-06	28	1	3.57
7	2026-01-07	28	1	3.57
8	2026-01-08	28	1	3.57
9	2026-01-09	28	1	3.57
10	2026-01-10	28	1	3.57
11	2026-01-11	28	1	3.57
12	2026-01-12	28	1	3.57
13	2026-01-13	28	1	3.57
14	2026-01-14	28	1	3.57
15	2026-01-15	28	1	3.57
16	2026-01-16	28	1	3.57
17	2026-01-17	28	1	3.57
18	2026-01-18	28	1	3.57
19	2026-01-19	28	1	3.57

Quastions .s...\Admin (54))* | LAST Q.sql - not connected

```sql
-- ============================================================
-- Q4: Reservation Details View
-- ============================================================

CREATE VIEW vw_ReservationDetails AS
SELECT
    r.ReservationID,
    g.FullName                                  AS GuestName,
    ro.RoomNumber,
    rt.TypeName                                 AS RoomType,
    r.check_in_date                             AS CheckInDate,
    r.check_out_date                            AS CheckOutDate,
    r.nights                                    AS Nights,

    -- Human-readable status label
    CASE r.reservation_status
        WHEN 'confirmed'    THEN 'Booked'
        WHEN 'checked_in'   THEN 'Checked-in'
        WHEN 'checked_out'  THEN 'Checked-out'
        WHEN 'cancelled'    THEN 'Cancelled'
        WHEN 'no_show'      THEN 'No Show'
```

110 % | 10 | 0 | Ln: 676, Ch: 1 | (2 chars, 1 lines) | MIXED | CRLF | UTF-8

Results | Messages

	ReservationID	GuestName	RoomNumber	RoomType	CheckInDate	CheckOutDate	Nights	ReservationStatus	TotalRoomCharge	BookingSource	NightlyRateAtBooking
1	49	Ali Omar	104	Double	2026-02-25	2026-03-05	8	Checked-in	7200.00	NULL	900.00
2	52	Ziad S	201	Single	2026-02-28	2026-03-04	4	Checked-in	2000.00	NULL	500.00
3	55	Karem A	209	Double	2026-02-28	2026-03-05	5	Checked-in	4500.00	NULL	900.00
4	58	Huda S	402	Single	2026-03-01	2026-03-05	4	Checked-in	2000.00	NULL	500.00
5	65	Wael H	302	Single	2026-03-01	2026-03-05	4	Checked-in	2000.00	NULL	500.00
6	72	Ola A	108	Double	2026-03-01	2026-03-05	4	Checked-in	3600.00	NULL	900.00

Parse (Ctrl+F5) | Admin (54))* | LAST Q.sql - not connected

```sql
-- ============================================================
-- Q5: Cancellation Analysis
-- ============================================================

-- Monthly cancellation performance
SELECT
    FORMAT(r.booking_date, 'yyyy-MM')           AS YearMonth,
    COUNT(*)                                    AS TotalReservations,
    SUM(CASE WHEN r.reservation_status = 'cancelled' THEN 1 ELSE 0 END)
                                                AS CancelledReservations,
    CAST(
        SUM(CASE WHEN r.reservation_status = 'cancelled' THEN 1.0 ELSE 0 END)
        * 100.0 / NULLIF(COUNT(*), 0)
    AS DECIMAL(5,2))                            AS [CancellationRate%]
FROM Reservations r
GROUP BY FORMAT(r.booking_date, 'yyyy-MM')
ORDER BY YearMonth;
GO

-- Top 5 cancellation reasons
SELECT TOP 5
```

110 % | 10 | 0 | Ln: 736, Ch: 3 | MIXED | CRLF | UTF-8

Results | Messages

	YearMonth	TotalReservations	CancelledReservations	CancellationRate%
1	2025-10	1	0	0.00
2	2025-11	4	0	0.00
3	2025-12	4	0	0.00
4	2026-01	9	1	11.11
5	2026-02	10	0	0.00

	CancellationReason	CancellationCount
1	No reason provided	1

Quastions .s...\Admin (54))* | LAST Q.sql - not connected

D:\Data analysis\Data mining\Graduation project\Quastions .sql - localhost.HotelReservation_BillingManagement (DESKTOP-BVV10T0\Admin (54))*

```sql
-- ============================================================
-- Q6: Services Revenue View
-- ============================================================

CREATE VIEW vw_ServiceRevenue AS
SELECT
    s.ServiceName,
    FORMAT(su.UsageDate, 'yyyy-MM')   AS YearMonth,
    COUNT(su.UsageID)                 AS TotalUsageCount,
    SUM(su.TotalCharge)               AS TotalServiceRevenue
FROM ServiceUsages su
JOIN Services s ON su.ServiceID = s.ServiceID
GROUP BY
    s.ServiceName,
    FORMAT(su.UsageDate, 'yyyy-MM');
GO

-- Query: Top 5 services by revenue in the last 3 months
SELECT TOP 5
    ServiceName,
    SUM(TotalUsageCount)    AS UsageCount,
```

110 % | 10 | 0 | Ln: 749, Ch: 1 | (2 chars, 1 lines) | MIXED | CRLF | UTF-8

Results | Messages

	ServiceName	UsageCount	Revenue
1	Spa	10	5500.00
2	Laundry	10	2300.00
3	Dinner	5	2100.00

Quastions .s...\Admin (54))* | LAST Q.sql - not connected

lose (Shift+Esc)

```sql
-- ============================================================
-- Q7: Invoice Aging & Outstanding Balances View
-- ============================================================

CREATE VIEW vw_InvoiceAging AS
SELECT
    i.InvoiceID,
    g.FullName                                      AS GuestName,
    i.InvoiceDate,
    i.NetTotalAmount                                AS TotalAmount,
    i.PaidAmount,
    (i.NetTotalAmount - i.PaidAmount)               AS OutstandingAmount,

    -- Age in days from invoice date to today
    DATEDIFF(day, i.InvoiceDate, CAST(GETDATE() AS DATE)) AS AgeDays,

    -- Aging bucket
    CASE
        WHEN DATEDIFF(day, i.InvoiceDate, CAST(GETDATE() AS DATE)) BETWEEN 0  AND 7  THEN '0-7 days'
        WHEN DATEDIFF(day, i.InvoiceDate, CAST(GETDATE() AS DATE)) BETWEEN 8  AND 30 THEN '8-30 days'
        WHEN DATEDIFF(day, i.InvoiceDate, CAST(GETDATE() AS DATE)) BETWEEN 31 AND 60 THEN '31-60 days'
```

110 % | 10 | 0 | Ln: 781, Ch: 1 | MIXED | CRLF | UTF-8

Messages

```
Commands completed successfully.

Completion time: 2026-03-17T11:17:42.1319501+02:00
```

110 % | No issues found | Ln: 4, Ch: 1 | TABS | MIXED | UTF-8 with BOM

Quastions .s...\Admin (54))* | LAST Q.sql - not connected

```sql
-- ============================================================
-- Q8: Payment Method Breakdown (Monthly)
-- ============================================================

SELECT
    FORMAT(p.PaymentDate, 'yyyy-MM')              AS YearMonth,

    SUM(CASE WHEN p.PaymentStatus = 'Completed'
             THEN p.AmountPaid ELSE 0 END)         AS TotalPaid,

    SUM(CASE WHEN p.PaymentMethod = 'Cash'
              AND p.PaymentStatus = 'Completed'
             THEN p.AmountPaid ELSE 0 END)         AS PaidByCash,

    SUM(CASE WHEN p.PaymentMethod = 'Card'
              AND p.PaymentStatus = 'Completed'
             THEN p.AmountPaid ELSE 0 END)         AS PaidByCard,

    SUM(CASE WHEN p.PaymentMethod = 'Online'
              AND p.PaymentStatus = 'Completed'
             THEN p.AmountPaid ELSE 0 END)         AS PaidOnline,
```

110 % | 10 | 0 | Ln: 827, Ch: 1 | (2 chars, 1 lines) | MIXED | CRLF | UTF-8

Results | Messages

	YearMonth	TotalPaid	PaidByCash	PaidByCard	PaidOnline	PaidByWallet	FailedTransactions	PendingTransactions
1	2025-11	10800.00	8300.00	2500.00	0.00	0.00	0	0
2	2025-12	8950.00	7400.00	1550.00	0.00	0.00	0	0
3	2026-01	33850.00	200.00	14400.00	12450.00	6800.00	0	0
4	2026-02	12400.00	0.00	2400.00	10000.00	0.00	0	0
5	2026-03	12300.00	9000.00	2300.00	0.00	1000.00	0	0

Quastions .s...\Admin (54))* | LAST Q.sql - not connected

D:\Data analysis\Data mining\Graduation project\Quastions .sql - localhost.HotelReservation_BillingManagement (DESKTOP-BVV10T0\Admin (54))*

```sql
-- ============================================================
-- Q9: Housekeeping & Room Turnover Report
-- ============================================================

-- View: housekeeping performance per room
CREATE VIEW vw_HousekeepingPerformance AS
SELECT
    h.RoomID,
    ro.RoomNumber,
    rt.TypeName                                     AS RoomType,
    ro.BuildingID,

    -- Logs in the last 30 days only
    COUNT(CASE
        WHEN h.CleaningDate >= CAST(DATEADD(day,-30,GETDATE()) AS DATE)
        THEN h.LogID
    END)                                            AS CleaningCount_Last30Days,

    -- Average cleaning duration in minutes (all time)
    AVG(DATEDIFF(minute, h.StartTime, h.EndTime))   AS AvgCleaningMinutes,
```

110 % | 10 | 0 | Ln: 866, Ch: 43 | MIXED | CRLF | UTF-8

Results | Messages

	RoomID	RoomNumber	RoomType	BuildingID	CleaningCount_Last30Days	AvgCleaningMinutes	LateCleaningSessions
1	R12	202	Single	B2	1	60	1
2	R13	203	Double	B2	1	60	1
3	R26	401	Single	B2	1	75	1
4	R27	402	Single	B2	1	45	1
5	R28	403	Double	B2	1	45	0
6	R29	404	Double	B2	1	45	0
7	R3	103	Double	B1	1	60	0
8	R30	405	Suite	B2	1	55	0
9	R4	104	Double	B1	1	40	0
10	R5	105	Suite	B1	1	60	0
11	R6	106	Single	B1	1	45	0
12	R7	107	Single	B1	1	45	0
13	R8	108	Double	B1	1	45	0
14	R9	109	Double	B1	1	60	0
15	R14	204	Double	B2	1	60	0
16	R15	205	Suite	B2	1	45	0
17	R16	206	Single	B2	1	45	0
18	R17	207	Single	B2	1	60	0
19	R18	208	Double	B2	1	75	0

Quastions .s...\Admin (54))* | LAST Q.sql - not connected

```sql
-- ============================================================
-- Q10: Staff Performance View (Last 30 Days)
-- ============================================================

CREATE VIEW vw_StaffPerformance AS
SELECT
    e.EmployeeID,
    e.EmpName                                       AS StaffName,
    e.JobRole                                       AS Role,

    -- Reservations this employee created/handled
    COUNT(DISTINCT r.ReservationID)                 AS ReservationsHandled,

    -- Check-ins processed (employee recorded as CheckedInByEmployeeID)
    COUNT(DISTINCT CASE
        WHEN r.CheckedInByEmployeeID = e.EmployeeID
        THEN r.ReservationID
    END)                                            AS CheckInsProcessed,

    -- Check-outs processed
```

110 % | 10 | 0 | Ln: 919, Ch: 64 | MIXED | CRLF | UTF-8

Results | Messages

	StaffName	Role	ReservationsHandled	CheckInsProcessed	CheckOutsProcessed	TotalPaymentsProcessed	TotalRevenueProcessed
1	Samy Galal	Billing Manager	0	0	0	2	7500.00
2	Samy Badr	Billing Manager	0	0	0	1	4200.00
3	Omar Saad	Billing Manager	0	0	0	1	2300.00
4	Mona Zaki	Billing Manager	0	0	0	1	2300.00
5	Ehab Galal	Billing Manager	0	0	0	1	1000.00
6	Sara Ahmed	Receptionist	3	0	0	0	0.00
7	Ahmed Ali	Receptionist	4	0	0	0	0.00
8	Fady Aly	Receptionist	0	0	0	0	0.00
9	Maya Ihab	Admin	0	0	0	0	0.00
10	Karem Gad	Receptionist	0	0	0	0	0.00